SYSTEMS REVIEW · THRIVE ALPHA

Decision-Support Systems Review

A working audit of the ticker detail page against a 10-second decision bar, plus what the backend actually feeds it — and a grounded path from where this already is (further along than it gets credit for) to serving scout, swing, and owner traders at institutional depth.

Scope ticker detail UX · gating accuracy · architecture · roadmap

Basis live code + 106 real EOD snapshots (2026-03-07 → 2026-08-25) **Date** 2026-08-25

00
Why this matters first
A bug already found and fixed

01
Ticker detail audit
Section by section, 10-sec bar

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Architecture read
What's strong, what's thin

03
Path to institution-grade
Prioritized, not aspirational

00 Why this matters before the UI does

A page can be beautifully organized and still lie to the person reading it. That happened here, live, and it's the reason the rest of this review leads with distrust of anything the model shows until traced to source.

Valuation gate had a 0% pass rate for 5+ months — fixed this session

FIXED

A display-only width cap meant to keep the long-term fair-value band readable in the UI (`_cap_band_width_for_display`, ±35% of spot) was silently feeding the live **valuation gate** and the TRIM **extension gate**. Checked against all 6,534

real LT-band observations in the download: **100.0%** sat exactly at the cap.
Result: `full_go` fired **zero times** across 106 days and 121 tickers — including bonds (TLT, HYG) that should never fail a value screen this hard.

`target_bands.py → retail_action_payload.py::_assessment_from_lt_band · trim_gates.py (same call)`

→ Uncapped fair-value midpoint now computed separately (`lt_fair_value_mid`) and fed to both gates; the capped band still drives the display. Verified: −14.8% → −4.1% on a synthetic case; legacy rows fall back safely.

The takeaway isn't "one bug, fixed." It's that **display-shaping helpers are not safe inputs to decision logic** by default in this codebase — and that's a pattern worth auditing once, deliberately, rather than finding one instance at a time. See §03 · P0.

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01 Ticker detail page, against the 10-second bar

The bar: open the page, and within 10 seconds state — act/wait/trim, where price matters, what proves you wrong. That's the product's own north star (`PRODUCT_THESIS.md`), not one I'm importing.

B−
10-SEC BAR

The answer *exists* and is well-built — `TickerDecisionHero` genuinely packs verdict, reason, levels, and gates into one scannable card. But it's not the first thing on screen, and by the time it appears the reader has already parsed two near-duplicate verdicts.

Fix the stacking order and the duplication, and this moves to an A without touching the model.

What's actually stacked before the answer

On first paint, before any tab content: header (ticker · price), an optional catalyst banner, an optional post-earnings banner, then `StickyTickerQuoteBar` (price, score, setup quality, regime, action title, gates X/Y, meta-confidence), a position-intent chip, and a mode selector — *then* the tab bar, *then* `TickerDecisionHero` repeats price, verdict

mode selector — *then* the tab bar, *then* `TickerDecisionHero` repeats price, verdict, and gates X/Y in a different visual encoding.

Verdict shown twice, unscrolled, in different shapes

MODERATE

Score, regime, action title, and the gates progress count appear in the sticky bar and are then substantially repeated in the hero card immediately below it. Neither is wrong; together they cost the reader a second parse before they're sure which one is "the" answer.

`tradingPlatformUX.tsx::StickyTickerQuoteBar + tickerDetail/TickerDecisionHero.tsx`

→ Slim the sticky bar to identity only (ticker, price, daily move) once the hero is on screen; let the hero own verdict + gates as the single source.

Four components independently render the same verdict, patched apart with hide-flags

STRUCTURAL

`StickyTickerQuoteBar`, `TickerDecisionHero`, `DecisionDeck`, and `InvestorDecisionSummarySection` each know how to draw a verdict/gates block. They're kept from colliding today via `hideHero`, `hideTradingGates`, `hideRoadmaps`, `compactOverview` props threaded through call sites — a sign of accretion, not a finished consolidation. `TickerDecisionHero`'s own header comment confirms it: it already replaced five earlier one-off verdict cards.

`ReferenceDetailGrid.tsx::ChartLevelsCard passes hideHero + hideTradingGates + hideRoadmaps + compactOverview in one call`

→ Worth finishing that consolidation: one verdict component, imported everywhere, no suppression flags.

A debug panel renders by default in the retail-facing valuation view

MODERATE

`FairValueDebugPanel` (TTM/forward P/E used, fair-value metadata) sits inline in the Why → Valuation sub-tab alongside `UnifiedValuationSummary`, with no toggle. It's genuinely useful for verifying the model — not for a retail trader trying to decide in 10 seconds.

`TickerDetailModal.tsx, whySubTab === "valuation"`

→ Move behind an explicit "Show model internals" disclosure, collapsed by default.

Internal file paths leak into retail-facing copy

POLISH

Empty-thesis state reads: *"No written thesis yet. Add in data/theses.yaml or via add_thesis.py"* — meaningless and slightly alarming to a retail user with no filesystem access.

`TickerDetailModal.tsx, Why → Context tab`

→ Rewrite as a null state ("No written thesis for this ticker yet") or hide the row entirely for non-holdings.

Dead component still shipping in the bundle

CLEANUP

`SignalReasonCard` the component has zero render call-sites left — only its exported types are still imported (`TickerDecisionHero`, `InvestorDecisionSummary`). 139 lines of unused UI code, artifact of the same consolidation that produced the hero.

`tickerDetail/SignalReasonCard.tsx`

→ Move the type exports to a types-only file; delete the component function.

What's already right — worth not disturbing

Tab split (Today / Levels / Why) and progressive disclosure

KEEP

Advanced material — terms glossary, sector-specific stats, rate sensitivity, tier ladders, drawdown/volume detail — is correctly collapsed or pushed to `Why`. `ReferenceDetailGrid`'s "one card per topic, second look" framing is a genuinely good pattern and explicitly designed against duplication (its own copy says so: *"one place — not duplicated across tabs," "correctly placed last"*).

Net read: nothing here needs a rebuild. The hero card is the right shape; the fix is subtraction — flatten the pre-hero stack, finish the hero consolidation the codebase already started, and stop showing internal state to end users. That's a few days of focused work, not a redesign.

02 Architecture, read honestly

Grounded in `CLAUDE.md`, `docs/planning/PRODUCT_THESIS.md`, and `TODO.md`'s living backlog — not a from-scratch opinion.

Further along than "add a feature" framing suggests

CAPABILITY	STATE
Multi-horizon signal engine	ST/MT/LT bands, ATR stops, regime from HMM, LightGBM per horizon per ticker — real modeling, not a single smart-score
4-gate discipline	Valuation · regime · momentum · ML, tiered by tranche depth — the right shape even where I found the valuation feed broken
Forward tracking + paper book	Signal-vs-outcome by horizon, entry-mode, gate combination; a real trust surface, not a marketing stat
Persona lens (Scout / Swing / Owner)	Phases A–E in `TODO.md` are ~95% shipped — mode-aware overview, hunter/scanner, session strip, catalyst-to-signal, calibration by lens
Confidence fusion	7-layer model (regime, ML, options, fundamentals, news, etc.) surfaced with an explicit unlock sequence, not a black box
Degraded-mode handling	Feature flags when FMP/FRED partially fail — most retail tools just show stale data silently

The structural risk class this session surfaced

The valuation-gate bug wasn't a one-off typo — it was a **category**: a helper computed for presentation (`_cap_band_width_for_display`) got consumed by decision math with no boundary between the two. `gate_backtest.py`'s own docstring shows this has happened before and was fixed *once, in the backtest path only* — the live path regressed independently. That's worth a deliberate, one-time audit: every `_cap_*_for_display`, rounding, or "just for the UI" helper, grep'd for callers outside display code.

The gap that actually blocks "serves all retail traders"

No multi-tenant identity — the app is still single-operator

BLOCKING

Portfolio holdings live in `config.yaml`. There is no sign-up, no per-user session, no portfolio import, no guest/authenticated split — all explicitly open items in `TODO.md`. Every other gap in this document is a refinement; this one is the difference between a personal tool and a product other people can use.

03 Path to institution-grade, for scout / swing / owner

Sequenced by what unblocks the next thing, not by what's most exciting to build.

P0 Fix trust leaks before adding surface area

Days, not weeks — protects everything below it

- **Audit display-cap → gate feeds** — the same class of bug as the valuation gate, checked everywhere it could recur.
- **Ticker detail consolidation** — one verdict component, slim sticky bar, debug panel behind a toggle, retail-safe copy (§01).
- **Confirm the fix live** — watch the next few `run_daily.py` runs on Railway; `full_go` should stop being permanently zero.

P1 Multi-tenant — turn the tool into a product

Unlocks every persona beyond the operator

- **Auth** — email or OAuth, session on the API.
- **Per-user portfolio store** — replace config-file holdings; `config.yaml` becomes the demo/guest default only.
- **Guest vs. authenticated shell** — anonymous gets the demo universe; signed-in gets their own book.
- **Portfolio import** — CSV / broker export, tied to login.

P2 Deepen the "institution-grade" claim

Where the quant substance catches up to the pitch

- **MCTR** — marginal contribution to risk per holding, not just concentration %.
- **Named historical stress scenarios** — 2008, COVID Mar 2020, 2022 rate hike, beta-adjusted per holding.
- **Liquidity score** — ADV-based days-to-exit; the Signals table placeholder column already expects this.
- **Redis OHLCV cache + worker dyno** — TODO.md Phase 2/3 scaling, needed before 150-ticker coverage is real.

P3 Expand how traders reach the decision

High-visibility, correctly sequenced last

- **Alerting** — email/push on regime flip, gate clear, confidence threshold; still not started.
- **LLM co-pilot** — structured context (gates, bands, confidence) into a chat surface, not a free-floating chatbot.
- **Execution assist** — broker order prefill from the copy-trade-plan data already generated.